

Weapon Identification Across Varying Acoustic Conditions Using an Exemplar Embedding Approach

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ABSTRACT

Gunshot recordings have the potential for both tactical detection and forensic evaluation particularly to ascertain information about the type of firearm and ammunition used. Perhaps the most significant challenge to such an analysis is the effect of recording conditions on the audio signature of recorded data. In this paper we present a first study of using an exemplar embedding approach to automatically detect and classify firearm type across different recording conditions. We demonstrate that a small number of exemplars can span the space of gunshot audio signatures and that this optimal set can be obtained using a wrapper function. By projecting a given gunshot to the subspace spanned by the exemplar set a distance measure/feature vector is obtained that enables comparisons across recording conditions. We also investigate the use of a hierarchy of gunshot classifications that assists in improving finer level classification by pruning out gunshot labeling that is inconsistent with its higher level type. The embedding based approach can thus be used both by itself and as a pruning stage for other search techniques.

Our dataset includes 20 different gun types captured in a number of different conditions. This data acts as our original exemplar set. The dataset also includes 12 gun types each with multiple shots recorded in the same conditions as the exemplar set. This second set provides our training and testing sets. We show that we can reduce our exemplar space from 20 to only 4 uniquely different gunshots without significantly limiting the ability of our embedding approach to discriminate different gunshots in the training and testing sets. The basic hypothesis in the embedding approach is that the relationship between the set of exemplars and space of gunshots including the testing/training set would be robust to a change in recording conditions or the environment. That is to say the embedding distance between a particular gunshot and the exemplars would tend to remain the same in changing environments. The implication of this are two-fold; first, unlike other dimensionality reduction approaches we have access to particular instances/examples of entities (the exemplars), which act as bridges to connect different recording conditions. Second, the embedding distances are invariant across recording conditions, the embedded vector can be used as a feature of similarity between gunshots recorded in different conditions.

Unlike other dimensionality reduction approaches, our approach generates descriptions that are always in terms of the same exemplars. In other approaches such as PCA, the data driven nature makes it difficult if not impossible to make correspondence in the dimensions in one space to another.

We have shown that gunshot classification across different recording conditions can be performed at a reasonable degree of certainty (60-72%) at a finer level (gunshot to weapon model) and at a high degree of certainty (95-100%) at a higher degree of abstraction (gunshot to 'handgun' or 'rifle'). We also investigate the use of simulated recording conditions and artificial noise to quantitatively evaluate the performance of our approach.

1 Introduction: Background and Motivation

Past work in audio classification has been carried out to classify broad categories such as speech, music, cheering etc. [2, 3] mostly using Gaussian Mixture Models (GMM's) and Hidden Markov Models (HMM's). Such broad categorization has sufficed for audio-visual event detection for applications such as consumer video browsing and surveillance. However, it falls short when a finer characterization of the event is required. Gunshot analysis (see [1] for a survey of the state of the art) requires fine classification of gunshots into precise weapon categories. Clavel et al [5] come closest to such fine classification since they detect and classify gunshots using a collection of sub-classifiers for guns, grenades etc. Other prior work in gunshot analysis by Freytag et al [4] and others [6-11] has been based on a non-hierarchical template matching over various weapons. The main disadvantage of the non-hierarchical approach is that it is time consuming because it is not possible to get any characterization of a given acoustic signature without going through the entire database of weapons. Secondly these approaches require the acoustic capture conditions to be consistent across training and testing gunshot samples. This constraint limits the applicability of weapon identification to controlled laboratory conditions or preselected environmental conditions. Circumventing this problem requires a canonical space of weapon signatures that can act as the bridge between different recording conditions and that is favorable to a hierarchical course to fine analysis of weapon acoustic signatures.

In this paper we present an exemplar embedding approach that shows that a relatively small number of exemplars, obtained using a wrapper function, can span an expansive space of gunshot audio signatures. By projecting/embedding a given gunshot into the exemplar space a distance measure/feature vector is obtained that describes the given gunshot in terms of the exemplars. The basic hypothesis in the exemplar embedding approach is that the relationship between the set of exemplars and space of gunshots including the testing/training set would be robust to a change in recording conditions or the environment. That is to say the embedding distance between a particular gunshot and the exemplars would tend to remain the same in changing environments. The implication of this are two-fold; first, unlike other dimensionality reduction approaches we have access to particular instances/examples of entities (the exemplars), which act as bridges to connect different recording conditions. Second, the embedding distances are invariant across recording conditions; the embedded vector can be used as a feature of similarity between gunshots recorded in different conditions.

We also investigate the use of a hierarchy of gunshot classifications that assists in improving finer level classification by pruning out gunshot labeling that is inconsistent with its higher level type. . The first level of hierarchy consists of classification into broad weapons categories such as Rifle, Hand-Gun etc. The second level of the hierarchy consists of classification into specific weapons such as 9mm, 357 etc. The embedding based approach can thus be used both by itself and as a pruning stage for other search techniques.

Though exemplar embedding is not a new concept, a similar concept was applied for visual object classification [15], this is the first application of this idea to the acoustic classification domain to our knowledge. Unlike other dimensionality reduction approaches, our approach generates descriptions that are always in terms of the same exemplars. In other approaches such as PCA, the data driven nature makes it difficult if not impossible to make correspondence in the dimensions in one space to another.

The rest of the paper is organized as follows. Section 2 describes the difficulty caused by variation in capture conditions. Section 3 describes our exemplar embedding approach and the hierarchical classification of weapon acoustic signatures. We present our experimental results in section 4 and discuss concluding remarks with future directions in section 5.

2 Difficulty Caused by Variation in Capture Conditions

There are several factors stemming from the mechanics of the gun as well as the capture conditions. The muzzle blast is the primary sound for sub-sonic bullets, and is influenced by the ammunition, gun barrel length (among other factors), as well as the presence of acoustic suppressors that disguise the weapon. The mechanical action is picked up only if the microphone is close to the weapon. For supersonic bullets, the shock wave precedes the muzzle blast and is comparably strong in signal power. For supersonic bullets therefore, even a single bullet will produce pairs of sounds. Propagation through the ground or other solid surfaces is mostly relevant when the recording device is close to the weapon. The speed of sound can be 5 times higher in solid media than in the air.

Lossy propagation and reflection – Variations in temperature, humidity, ground surfaces and obstacles directly influence the extent of attenuation and scattering. The wind direction can affect the perceived frequency of the gunshot. These effects are not significant at a distance of 25 meters but become noticeable at a distance of 100 meters or greater. Finally, the angle between the gun and the microphone also plays a role especially since the microphone has a directional characteristic. We propose a data-driven approach based on robust signal processing for feature extraction and machine learning, since we cannot physically model all possible variations of the factors mentioned above. Our approach relies on training classifiers derived by using machine learning to classify weapon shots with robust features extracted from the training data and the actual test data. The advantage of this approach is that we can capture a wide range of operating conditions by capturing appropriate data in realistic conditions. Complex non-linear models underlying the data can be implicitly represented in terms of the classifiers. Furthermore, this approach enables incrementally adding new weapon types as more data becomes available, as well as adding more diversity of weapon sounds for those already in the database. Another important aspect of our data driven and classification approach is that we can enable similarity matching to a large database of already captured sounds for retrieving similar/same weapons from a large collection.

Effects of variability in recording devices - Freytag et al show that [4] the same weapon with the same ammunition yields significantly different signatures for each recording device. As Maher [1] points out, gunshots are impulse like signals and therefore the signatures are as informative of the overall capture conditions as they are of the nature of the gunshot. In other words, once again, we can account for small changes in the recording device but will have to depend on a family of models trained on a suitable variety of recording devices, with a model for each recording device. If we choose a wide enough variety of recording devices, we expect to find at least one recording device that is acceptably close to the capture device used by the query signature, and thus find a matching weapon.

3 Exemplar based Embedding

The central hypothesis in our approach is that the space of gunshot acoustic signatures can be modeled as a subspace spanned by a minimal set of gunshot types or “exemplars”. Given these exemplars a test acoustic signature can be projected or “embedded” into the exemplar subspace creating a unique descriptor that can be used for gunshot detection, gun type classification tasks amongst others. In the following we describe our process of feature extraction from gunshot acoustic signatures, the exemplar embedding, exemplar selection approach and finally the classification approach using exemplar embedding.

3.1 Feature Selection & Exemplar Descriptors

Feature extraction is performed using a 30ms sliding window (10ms overlap) over the gunshot time duration and computing 13 Mel Frequency Cepstral Coefficients (MFCCs). We empirically determined the expected time duration of gunshots to be 0.5s based on SNR. Each frame is then multiplied by a hamming window function:

$$w_i = (0.5 - 0.46(\cos(2\pi/N))), \quad 1 \leq i \leq N,$$

where N is the number of samples in the window. After performing FFT on each windowed frame, MFCC (Mel-Frequency Cepstral Coefficients) are calculated using the following Discrete Cosine Transform:

$$C_n = \sqrt{\frac{2}{K}} \sum_{i=1}^K \log S_i \times \cos(n(i - \frac{1}{2})\pi/K), n = 1, 2, \dots, L$$

where K is the number of sub bands and L is the desired length of cepstrum. $S_i, 1 \leq i \leq K$, represent the filter bank energy after the passing through the triangular band pass filters. The band edges for these band pass filters correspond to the Mel frequency scale (linear scale below 1 kHz and logarithmic scale above 1 kHz). We select the first thirteen coefficients resulting in a 13 dimensional feature vector associated with a given gunshot acoustic signature.

By “Exemplars” we mean a set of representative gunshot types that have the potential to span the entire space of gunshot types in our domain. In other words we hypothesize that each gunshot type can be represented in terms of varying degrees of affinity to the gun types in the exemplar set. For each of our gunshot exemplars E_i we train a Gaussian Mixture Model G_i on a set of MFCC feature vectors obtained from a number of gunshot examples of the respective gun type (For details on GMM’s and MFCC extraction, please see [2]). These act as the descriptors for each exemplar and provide a means to obtain the degree of affinity to another gunshot type.

3.2 Exemplar Embedding

As described earlier for each potential exemplar we use a set of training examples to generate a GMM from the MFCC extracted from their acoustic signatures. These GMMs serve as descriptors for each of the exemplars. Suppose there are N elements in our exemplar set. For each exemplar E_i a GMM descriptor G_i is learned from training examples. Thus we have a set of exemplar descriptors: $[G_1, G_2, \dots, G_N]$. Given a sufficiently expansive set of exemplars we can hypothesize that the exemplar descriptor set spans the space of gunshot acoustic signatures in our domain of interest. Now the act of exemplar embedding is to project a test acoustic signature into the space of exemplar descriptors. This is performed by obtaining the MFCC feature x_i of the test gunshot and obtaining the likelihood $l_i = G(x_i)$ that it belongs to the exemplar descriptor E_i . The result as shown in figure 1 is a feature vector $L = [l_1, l_2, \dots, l_N]$ that we call the embedding vector.

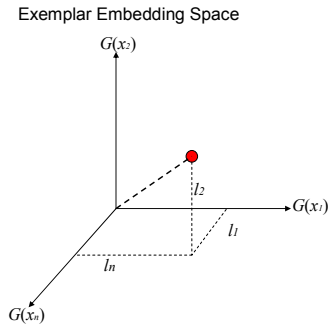


Figure 1: Exemplar embedding. A gunshot MFCC feature x_i is projected into the exemplar space by obtaining the likelihoods $l_i = G(x_i)$ for each of the exemplar descriptors.

3.3 Exemplar Selection

Identifying discriminative exemplars is an essential step of the learning process. Previous works use motion energy minima and maxima [11, 12], or k-means clustering (adapted to return exemplars) [13] to this end. However, there is no apparent relationship between such criteria and the discriminant quality of the selected exemplars. In light of this, we use a wrapper [14], a technique for discriminant feature subset selection. The idea behind a wrapper is to use the trained classifier itself to evaluate how discriminative a candidate set of exemplars is. Thus a wrapper performs a greedy search over the full set of exemplars where in each

iteration, classifiers are learned and evaluated for each possible subset considered. The wrapper method we use is called backward elimination [ref], and proceeds as follows.

Let E denote the initial set of exemplars. Given training gunshots:

1. Set $X = \emptyset$
2. Find $e \in E$, where k -means clustering of the training gunshots using $Y - y$ as embedding exemplars has best clustering performance.
3. Set $Y = Y - y$ and add $X = X \cup y$
4. Go to step 2 and repeat till $Y = \emptyset$.

The crucial step in the above algorithm is step 2 where we evaluate the ability of a *reduced* exemplar set to distinguish between a set of training gunshot examples. Our criteria for evaluating exemplar set's discriminatory power are quite straight forward. For each of the training gunshot examples we obtain the embedding vector L using the exemplar set. These embedding vectors are then clustered using k -means clustering. The clusters are evaluated for their accuracy by comparison with ground truth labels. In step 2 of the above algorithm we sequentially remove one of the exemplars in the exemplar set and compute the clustering accuracy of the reduced exemplar set. The exemplar that has the least effect on the clustering performance is permanently removed from the exemplar set. In this fashion at every iteration of the algorithm we prune the exemplar set and the best clustering performance is recorded. Figure 2 shows the plot of clustering accuracy over the training set for increasing iterations of the wrapper method. At each iteration, the exemplar with the least impact on clustering accuracy is removed. Our initial exemplar set consisted of 20 different gunshot descriptors all of which were generated from multiple gunshot acoustic signatures recorded in the same environmental conditions. The training set consisted of approximately 100 gunshot signatures randomly selected from different gun types in the exemplar set and separated prior to this experiment. As can be observed in figure 2(a), as we progress through the pruning of the exemplar set the clustering accuracy varies. An interesting thing to notice is that initially the clustering accuracy remains constant but after 5 of the exemplars are removed from the set, the clustering accuracy improves indicating that our original exemplar set not only had redundancy but also the fact that this redundancy can increase the complexity of the system to a level where inference tasks like k -means or other classification approaches may be confused. From iteration 6 to 16 we have another plateau in clustering performance. At this point any further reduction in the exemplar set results in a monotonically decreasing training set clustering accuracy. This suggests that four remaining exemplars (marked in red text in figure 2b) is the minimal set of exemplars that we must maintain to achieve a satisfactory level of discriminatory power from the embedding vectors. Therefore as a result of the pruning in the wrapper algorithm we have obtained a reduced set of exemplars that we shall use for embedding based classification.

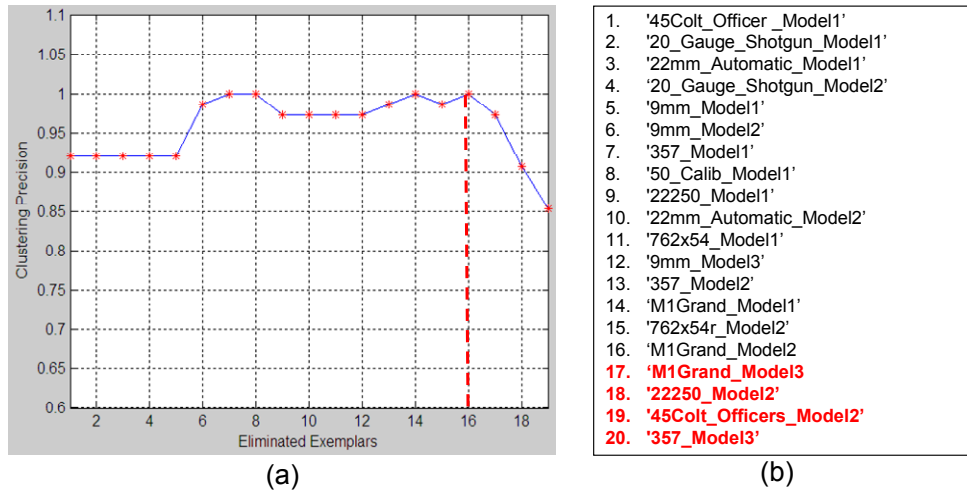


Figure 2: In (a) we show the clustering accuracy of the training set using sequentially reduced number of exemplars to generate the embedding vectors. (b) shows the set of gunshot types used as the exemplars. The exemplars are numbered according to the sequence in which they are eliminated from the exemplar set using the wrapper method. Exemplars in red are the ones we selected as our minimal set.

3.4 Application of Embedding to Capture Condition Invariant Weapon Identification

Using the exemplar set each new gunshot can be described as an embedding vector in the exemplar space i.e. in terms of likeness or affinity to each of the exemplars. As we shall see we can use this exemplar embedding vector as the underlying bridge to communicate between different capture conditions. Assuming that differing environmental conditions preserves the inherent relations between the different gunshot acoustic signatures we use the same exemplar set across varying acoustic capture conditions. For each capture condition a new set of descriptors are trained for the exemplars using gunshot examples obtained in the particular capture conditions. The result is a set of gunshot descriptors for each different capture condition using the same set of exemplars. This means the embedding vectors obtained from different capture conditions can communicate and interact in a single embedding space. Figure 3 illustrates this.

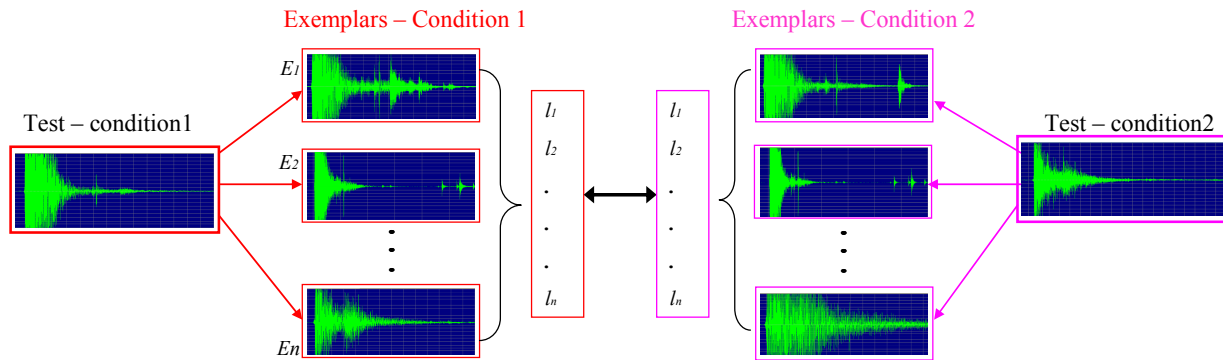


Figure 3: Exemplar embedding across different capture conditions. For each different capture condition the same gun types are used as exemplars and test gunshots are embedded using the same gun type exemplars. This allows comparison across capture conditions as the embedding vectors are in terms of the same exemplars.

For each gunshot sample in our training set we obtain the embedding vector using exemplars in the matching conditions. These embedding vectors are then clustered using k-means clustering and the cluster centroids of each gun type are used as descriptors for each gun class. In the classification stage, given a test gun shot sample we first project it into the exemplar embedding space using the exemplar set that best matches the

capture conditions of the test gun shot. Note we assume we already have prior knowledge of the test gun shot's capture conditions. The task of automatically determining the capture conditions from an acoustic signature is a challenging problem in itself and we do not in this paper propose a solution for this problem. Given the embedding vector for the test, gunshot classification is performed by computing its distance from each of the gunshot type cluster centroids and choosing the nearest class.

4 Experimental Results

To generate our exemplar set we used a pool of 20 different gunshots types all recorded in the same capture conditions (outdoors approx 10 m from source) which consisted of a variety of rifles and handguns like 45 Colt, 9mm, 50 Caliber, 20 Gauge Shotgun etc (see figure 2 for details). For training and testing we had a separate pool of gunshots consisting of between 5 to 15 samples of each gun type. The training set was used in our exemplar selection algorithm (section 3.3) to obtain a reduced set of 3 exemplars: M1Grand (rifle), 22250 (rifle), 45Colt (handgun) and 357 (handgun). The training set was also used to obtain the cluster centers for each gun type in the exemplar embedding space.

To test performance across recording conditions we simulated different capture conditions including: "Room Reverb", "Concert Reverb" & "Doppler Effect". Each of the exemplar and test gunshot sample was modified with the appropriate modulation. Exemplar embedding was performed in the respective capture conditions and embedding vectors were compared across conditions. A true classification was marked as one when a test gunshot sample from a different capture condition was classified or matched to the correct gun type class cluster in the original capture conditions. Table 1 shows the performance of our approach. Note that "In First 2", "In First 3" means the correct classification is amongst the two and three closest clusters respectively, whereas "First" means the correct classification is also the closest cluster.

	Room Reverb	Concert Reverb	Doppler
In First 3	0.99	0.93	0.71
In First 2	0.83	0.75	0.51
First	0.69	0.6	0.41
Handgun/Rifle	1	0.97	0.96

Table 1: Classification accuracy for embedding based approach for different capture conditions.

We also tested our approach on a reduced number of classes. Instead of all 20 gunshot types we divided the testing set into two classes: Rifle and Handgun. As can be seen in table 1, classification accuracy is much improved with this reduced number of classes. This suggests a hierarchy of gunshot classifications that can assist in improving finer level classification by pruning out gunshot labeling that is inconsistent with its higher level type. The embedding based approach can thus be used both by itself and as a pruning stage for other search techniques.

Conclusions and Future Directions

In this paper we have presented a first study of using an exemplar embedding approach to automatically detect and classify firearm type across different recording conditions. We demonstrate that a small number of exemplars can span the space of gunshot audio signatures and that minimal set can be obtained using a wrapper function. By projecting a given gunshot to the subspace spanned by the exemplar set a distance measure/feature vector is obtained that enables comparisons across recording conditions. The embedding based approach can be used both by itself and as a pruning stage for other search techniques.

In the future we plan to extend our approach to large databases of gunshot acoustic signatures as well as increasing the number of capture conditions. Another interesting direction for further study would be investigate the performance of our exemplar embedding approach to different categories of sounds like musical instruments, animal sounds and even speech recognition.

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